TESTNET · TEST KEYS

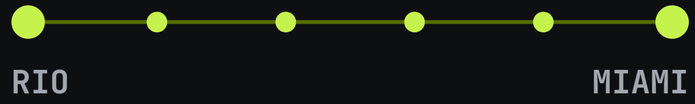
Radox Custody Rail

SWIFT for goods in transit, on Stellar

One shared custody record for goods in transit: every handover signed by both companies, checkable by anyone.

First corridor: Rio–Miami

Scan: the live shipment
rio-miami-rail.vercel.app



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PMF · THE PROBLEM

Who holds the whole chain?

Warehouse — Port — Carrier — Customs — Recipient

The gap: each company records only its own leg, and no shared record follows the goods.

45M

ocean bills of lading issued a year

DCSA, Feb 2023

~11%

of bills of lading issued electronically, 2025

DCSA, as of 2025

22%

of recorded cargo thefts in 2025 involved insiders

TT Club & BSI Cargo Theft Report, Apr 2026

Ports run legacy systems: over 90% of ports in low- and middle-income countries have no port community system (World Bank remark relayed by IAPH, Nov 2023); IMO guidance plans for legacy systems with proprietary formats (FAL.5/Circ.54, 2024).

Context: our explainer video on a Rio–Miami trade-corridor initiative; this project is its custody layer.

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PMF · WHAT IT COSTS

Cargo goes missing. No shared record.

The gap: a financier cannot see who holds the goods before paying.

R$900M

est. direct loss, 8,570 cargo robberies in Brazil, 2025

NTC&Logística survey, 28 Mar 2026

80–90%

of world trade relies on trade finance

WTO (undated page)

US$2.5T

global trade finance gap, unchanged from 2023

ADB Global Trade Finance Gap Survey, Dec 2025

These figures size the market problem the rail addresses. Sources under each number.

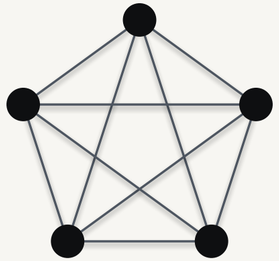
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THE PROPOSAL · A PUBLIC STANDARD

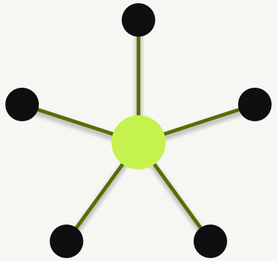
A SWIFT-like standard: integrate once

The goal is a shared, SWIFT-like custody standard that companies along a supply chain, from origin to destination, will use. **We are building it on Stellar, starting with the custody rail.**

PAIR BY PAIR · 10 LINKS



ONE STANDARD · 5 LINKS



Example, 5 companies: 10 custom links, or 5 links to one standard.

OPEN · ANYONE

— Record format, roles and events: the custody_registry contract (Apache-2.0)

— Verify any shipment, no login (slide 6)

RADOX WILL OPERATE · CLIENT PAYS

— Onboarding, issuance controls, operation of the rail

— The API and backend: closed, not self-hostable

— Per-institution subscription + fee per shipment (slide 9)

Each company integrates once instead of pair by pair. Radox is part of the Rio–Miami trade-corridor initiative and its technology rail: every port, airport and chamber that joins it will join the rail, through one adapter to the Radox API. Radox is in conversations with chambers of commerce in Brazil and Miami. Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

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THE RAIL · LIFECYCLE BUILT

One rail, origin to delivery

One record will follow the goods across company boundaries; port and airport systems will connect by API, not a rewrite.



— **Rule of every link:** both sides sign one Stellar transaction (delivery: recipient and holder), or the call fails.
— Each step extends the on-chain `head`: anyone can replay the history from Horizon and check it against the head.
— The token stays with its holder in transit; only custody moves.

Built: this lifecycle on testnet, test keys. Next: real companies on the rail, the standard, integrations.

Keyed by sha256(container | origin); sits beside the carrier's bill of lading. [1] Radox policy, not enforced on-chain; redeem needs the holder's signature.

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THE PROOF · LIVE ON TESTNET

The lifecycle ran on testnet

Warehouse → Port, both signed	ledger 5381142 · 635c99…f0bd38
Port → Carrier, both signed	ledger 5381143 · 56a0f0…7b7cc2
Event attested by Source	ledger 5381144 · 333792…068905
Carrier → Customs, both signed	ledger 5381145 · ac268a…c3c4b2
Customs → Recipient, both signed	ledger 5381146 · 5199d4…400b7a
Delivered by Recipient to Holder, both signed	ledger 5381147 · 919def…144e37
Redeemed: token burned	864fd1…15644d

Verify head

`2b3d51f0` open · operator
`635c99ac` handover 1 · Warehouse → Port
`56a0f006` handover 2 · Port → Carrier
`ac268a49` handover 3 · Carrier → Customs
`5199d4fb` handover 4 · Customs → Recipient
`919def8c` deliver · Recipient → Holder
`864fd164` burn · existing redeem

7 successful

9 refused, e.g. wrong signer (simulated)

24 unit tests

MATCH head check, 25 Oct

`custody_registry`
`CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII`

Also on testnet: handovers signed from separate computers (proofs/three-machine.json); two test companies' chains on one registry, sharing stops, each dual-signed (proofs/multi-chain.json).

Testnet, test keys.

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ECOSYSTEM · WHY STELLAR

Why Stellar: all signers, one call

01	**Multi-party auth in one call**	Soroban authorises both companies of a link inside one contract invocation: the handover is atomic.
02	**History from Horizon, checked by an on-chain head**	RPC keeps contract events about 7 days, so we replay each shipment's transactions from Horizon and check them against the head stored in the contract.
03	**Any Stellar asset**	The record stores a token address and never calls the token, so any tokenized Stellar asset can carry a custody record.

Stellar holds about US$4B in tokenized assets (SDF, as of 1 Sep 2026). **Nearest neighbours** (LumenLoop, SCF, Scout, 24–25 Oct): Kutana and Trustless Work release escrow on trade milestones; Pipeline finances commodity trade. None records a handover signed by two custodians; no SEP.

Open source, Apache-2.0: a custody primitive any Stellar tokenized-asset issuer can reuse. Radox will open API access to them, end to end.

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HACKATHON PROGRESS

Built this weekend, on disclosed prior work

PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform (closed backend); its 3 Soroban contracts are on Stellar mainnet, public since 2026-08-19

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Architecture notes (20 Oct), planning files (24–25 Oct), the explainer video, our agent tooling (from 22 Oct)

— No contract code before then: first repo commit 25 Oct 09:00 UTC

BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script, the public shipment page, this deck

— Radox API custody route reads the registry, read-only: same head (testnet)

AI attribution Code, tests, scripts, the shipment page, docs and this deck were made with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

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MARKET · MODEL · THE ASK

One rail, every corridor

MARKET · SOURCED FIGURES

US$94.2B

US–Brazil goods trade, 2025[1]

US$6.1B

Florida's goods exports to Brazil, its #1 market, 2025[2]

Global: US$26.26T world merchandise exports, 2025[3]; ~80% of goods trade by volume goes by sea[4]; ~45M ocean bills of lading a year[5].

MODEL

Subscription per connected institution + a fee per shipment.

Borrows the shape of Swift's billing rules: entry, annual, per-message usage[6]. Not affiliated with Swift.

Now running on testnet, test keys. **Next** the Rio–Miami pilot shipment, SCF Build, audit, mainnet, API for other Stellar assets.

[1] US Census Bureau, Trade in Goods with Brazil, 2025 (exports + imports, sum ours) [2] USTR, Florida trade data, 2025 [3] WTO, Mar 2026 [4] UNCTAD, Review of Maritime Transport (undated page, opened 25 Oct 2026) [5] DCSA, Feb 2023 [6] swift.com billing rules, accessed 25 Oct 2026

Our ask **Introductions to one financial institution and one port, airport or logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

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THE TEAM

Why us: a founder already on Stellar mainnet

Pedro Wakigawa Saragossy

Founder, captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

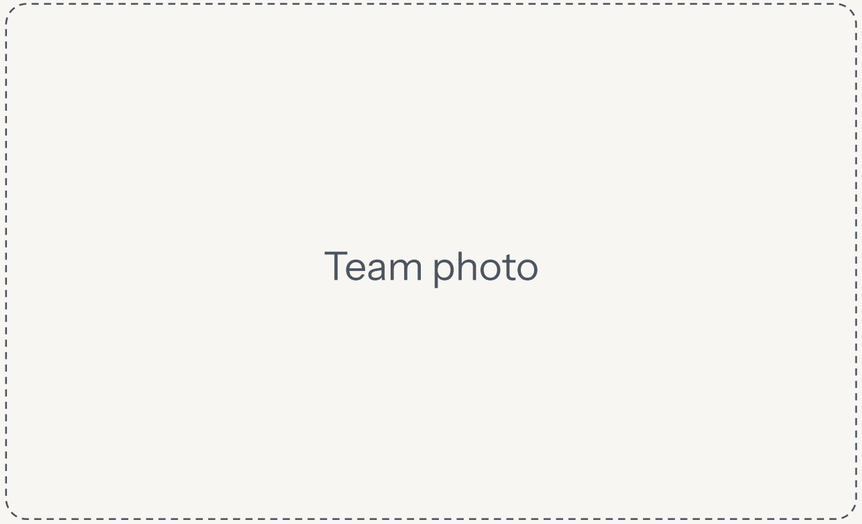
Leo Schlanger

Developer: custody_registry, the testnet harness and the proofs.

Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.



Rio–Miami first. Our goal: one rail for goods, every corridor.

Live page rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

AI Code, page, docs and deck made with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

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